

GEN-3 SAFE HELICOPTER

DOCUMENT 00

10-TON / 7-ROTOR SAFETY-CENTERED BASELINE

Fixed Safety Capsule Configuration

Conceptual Design Document • Revision 00-T10-R6

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PROPRIETARY CONCEPT INFORMATION

Concept / Pre-Feasibility Baseline — Not Certified

Baseline lock: *The earlier 100–200 ton concepts are superseded for this product baseline. This document establishes the new compact 10-ton architecture with seven **6.0 m** rotors, 1,600 L planning fuel capacity, and a fixed non-jettisonable safety capsule.*

1. Document Control

Item	Controlled Value
Document Number	00
Revision	00-T10-R6
Project	Gen-3 Safe Helicopter
Product Baseline	10 metric tonnes / 7-rotor compact platform
Rotor Configuration	7 rotors, nominal diameter 6.0 m each
Capsule Configuration	Fixed, non-jettisonable safety capsule
Fuel Planning Capacity	1,600 L nominal planning capacity
Overall Envelope	17 m × 17 m × 4 m target envelope
Founder	Amirbahador Atai
Status	Concept / pre-feasibility; engineering validation pending

This document is a design baseline, not a type certificate, flight manual, structural substantiation package, or production drawing set. Quantities identified as targets, planning values, or preliminary estimates must be closed through analysis, supplier data, component testing, system integration testing, and the applicable certification process.

2. Executive Summary

The project is deliberately shifted from the previous very-large conceptual vehicle family to a compact 10-ton rotorcraft intended to be easier to prototype, demonstrate, finance, and progressively validate. The defining product feature remains occupant survival through a dedicated safety architecture, but the safety concept is re-expressed in a manufacturable fixed-capsule configuration.

- 10,000 kg maximum design mass target.
- Seven primary lift rotors, each with a nominal **6.0 m** diameter.
- 1,600 L nominal fuel planning capacity.
- 17 m × 17 m × 4 m target overall envelope.
- Central fixed safety capsule, mechanically isolated from the main airframe.
- Multiple mechanical energy-absorption layers between rotorcraft structure, capsule support, and occupant zone.
- Fire detection, suppression, isolation, and controlled power reduction as independent safety functions.
- Distributed propulsion architecture designed so a single propulsion-unit failure does not automatically imply total loss of lift.
- Emergency descent architecture includes aircraft-level recovery provisions and dedicated capsule-support concepts, subject to certification and test validation.
- The architecture is intentionally modular so later versions may introduce a detachable capsule without requiring the entire platform architecture to be redesigned.

The document therefore treats the fixed-capsule version as the flagship demonstrator baseline, while preserving the safety philosophy and interfaces needed for a later detachable-capsule variant.

3. Frozen / Controlled Product Baseline

Parameter	Baseline	Status
Maximum design mass	10,000 kg target	Controlled target
Rotor count	7	Controlled
Rotor diameter	6.0 m nominal	Controlled
Total rotor disk area	197.9 m² theoretical geometric area	Calculated
Fuel capacity	1,600 L nominal planning capacity	Controlled target
Overall envelope	17 m × 17 m × 4 m	Controlled target
Safety capsule	Fixed / non-jettisonable	Controlled
Capsule isolation	Mechanical isolation + multiple damped load paths	Architecture
Primary safety philosophy	Layered containment, isolation, energy absorption, fault tolerance, emergency recovery	Controlled
Propulsion	Distributed; engine/fuel architecture not frozen in this revision	Open engineering item

4. Vehicle Architecture

The 10-ton platform is organized around a central safety capsule and a distributed lift system. The seven-rotor arrangement is treated as a system-of-systems rather than seven isolated propulsion units: each rotor station has mechanical, electrical, thermal, control, and structural interfaces.

- Central occupant capsule positioned close to the vehicle center of gravity.
- Primary airframe forms a load-carrying perimeter around the capsule.
- Rotor stations are distributed around the vehicle envelope to provide lift redundancy and control authority.
- Fuel and hot machinery are separated from the occupied capsule by dedicated containment and isolation zones.
- Landing and ground-impact loads are routed through sacrificial or energy-absorbing structures before reaching the capsule.
- The architecture prioritizes controlled graceful degradation rather than reliance on one heroic emergency device.

The compact architecture is intended to reduce development complexity relative to the superseded large concepts while retaining the distinctive safety-centered identity of the project.

5. Ten-Layer Safety Architecture

Layer 1 — Multi-sided structural safety cage

A reinforced structural cage surrounds the capsule and provides the primary load path around the occupant volume. The cage is intended to protect the capsule from direct intrusion while providing multiple load paths.

Layer 2 — Soft energy-absorbing isolators

Compliant isolators and high-stroke damped elements reduce the transmission of shock and vibration into the capsule.

Layer 3 — Controlled flexible capsule-to-airframe interfaces

Capsule attachment points are designed as controlled interfaces rather than rigid direct joints, allowing prescribed displacement and load filtering.

Layer 4 — Seven primary mechanical dampers

A distributed damper set under the main airframe provides a second mechanical energy-absorption stage between the vehicle structure and the capsule support system.

Layer 5 — Six capsule-side mechanical dampers

A second damper group beneath the capsule provides localized attenuation and load redistribution close to the occupant module.

Layer 6 — Fire detection, suppression, and isolation

Thermal sensing, fire detection, suppression, fuel isolation, and controlled power reduction are treated as separate safety functions.

Layer 7 — Fire containment around critical zones

Hot sections, fuel-system interfaces, and other high-consequence zones are separated from the capsule by dedicated barriers and containment provisions.

Layer 8 — Aircraft-level emergency recovery

A whole-aircraft emergency recovery system is retained as a concept-level additional survival layer. Parachute sizing, deployment dynamics, structural loads, and certification remain validation items.

Layer 9 — Capsule-support emergency descent

No under-capsule fans are included in this fixed-capsule baseline. Emergency descent provisions remain aircraft-level and subject to validation.

Layer 10 — Multi-stage occupant energy management

Seat, floor, capsule, dampers, and structural crush zones are treated as a combined energy-management chain rather than a single absorber.

Safety claim discipline: no layer is treated as a guaranteed survival device at this stage. The safety case must quantify failure probabilities, load cases, deployment envelopes, occupant injury criteria, and test evidence before certification credit is claimed.

6. Fixed Safety Capsule

The flagship 10-ton version uses a fixed, non-jettisonable safety capsule. The capsule remains mechanically integrated with the aircraft but is intentionally isolated from direct structural shock paths.

- Rigid occupant shell with controlled deformation philosophy.
- Multiple mechanical isolators between capsule and primary airframe.
- Independent internal energy-absorption provisions at the seat/floor interface.
- Dedicated emergency egress routes designed to remain functional after defined impact cases.
- Fire-resistant separation between occupied volume and propulsion/fuel zones.
- Water-survival provisions are considered part of the capsule architecture.
- Capsule systems are designed for controlled failure isolation where practical.

The fixed capsule is selected for the first demonstrator because it avoids the added mass, separation dynamics, certification complexity, and control problems associated with a detachable capsule. A future detachable version can reuse the capsule geometry and many of the safety interfaces as a separate product branch.

7. Seven-Rotor Propulsion Architecture

The seven primary rotors are the central lift system. Each rotor has a nominal **6.0 m** diameter. The preliminary geometric disk area is approximately **197.9 m²** for seven rotors combined.

Item	Preliminary value
Rotor count	7
Rotor diameter	6.0 m
Radius	3.0 m
Area per rotor	28.27 m²
Total geometric disk area	197.9 m²
Hover thrust per rotor at 10,000 kg gross mass	14.01 kN
Illustrative 20% total-thrust margin point	16.81 kN per rotor

Using ideal momentum theory only, the geometric rotor disk area gives an idealized hover induced-power floor of approximately **1.40 MW** at sea-level standard density and 10,000 kg gross mass. This is not installed shaft/electric power. Real installed power must include induced, profile, mechanical, transmission, electrical, control, atmospheric, and reserve margins.

The propulsion architecture therefore remains an engineering optimization problem. The preferred direction is a distributed power architecture in which multiple prime movers or generators feed independent propulsion channels, with sufficient control and power-management capability to maintain controllability after defined single-unit failures.

8. Preliminary Engine / Powerplant Selection

No specific commercial engine is frozen by Document 00. The correct selection must be made after the rotor aerodynamic model, mission profile, altitude, cruise speed, hover reserve, gearbox or electric-machine efficiency, thermal limits, and certification strategy are closed.

Architecture	Advantages	Primary concern
Multiple turboshafts + mechanical distribution	High power density; mature aviation architecture; direct shaft power	Transmission complexity and common-mode failure paths
Turbogenerator + distributed electric drives	Flexible packaging; independent rotor channels; fault isolation; scalable control	Generator, inverter, motor and thermal-management mass
Hybrid turbogenerator + battery buffer	Transient power support; smaller generator peak rating; redundancy potential	Battery mass, thermal management, certification complexity

Preliminary recommendation: investigate a turbogenerator-based distributed-electric architecture first, while retaining a multi-turboshaft mechanical architecture as a reference case. This is a recommendation for the engineering trade study, not a final engine selection.

9. Preliminary Mass Budget

The following is a planning-level mass allocation intended to keep the 10-ton baseline internally visible. It is not a certified weight statement. The allocation must be replaced by component-level supplier data as the design matures.

Mass group	Planning mass	Comment
Primary airframe and rotor support structure	2,050 kg	Includes major load paths and rotor support concept
Safety capsule and occupant protection structure	1,150 kg	Shell, isolators, dampers, seats and protection provisions
Seven rotors, hubs and pitch/control hardware	700 kg	Target allocation (6.0 m); supplier-dependent
Powerplant / generators / motors / drives	1,550 kg	Open trade-study item
Fuel system excluding fuel	220 kg	Tanks, pumps, plumbing, protection
Fuel, 1,600 L planning capacity	1,280 kg	Planning density 0.80 kg/L
Avionics, flight control and electrical distribution	520 kg	Includes redundancy allowance
Landing gear / ground interface	420 kg	Preliminary
Fire, emergency recovery and safety equipment	520 kg	Preliminary allocation
Thermal management / cooling	350 kg	Powerplant-dependent
Interconnects, hardware, fluids and growth reserve	1,290 kg	Development margin and unresolved equipment
Total planning mass	9,770 kg	Leaves ~230 kg to the 10,000 kg target

The 230 kg residual is intentionally not assigned to a passenger or payload claim. It is a design-closure reserve until the propulsion and structure selections are validated.

10. Fuel and Energy Architecture

The baseline includes 1,600 L of fuel capacity. Fuel is treated as a high-consequence energy source and therefore participates directly in the safety architecture.

- Segregated fuel containment away from the primary occupant capsule.
- Fuel shutoff capability following crash, fire, or severe system fault detection.
- Fire-resistant routing and protected lines through high-risk zones.
- Leak detection and pressure/flow monitoring.
- Controlled venting strategy.
- Thermal shielding between hot machinery and fuel compartments.
- Power management capable of reducing nonessential loads after failures.

At the planning density of 0.80 kg/L used only for the mass budget, 1,600 L corresponds to approximately 1,280 kg of fuel. Actual mass and usable fuel must be determined from the selected fuel specification and tank design.

11. Flight-Control and Fault-Reconfiguration Architecture

The seven-rotor layout is intended to support distributed control and graceful degradation. The flight-control system must detect abnormal rotor speed, thrust, temperature, vibration, power, and actuator behavior and then reconfigure the control law within validated limits.

- Independent monitoring of each rotor channel.
- Cross-checking of commanded versus measured rotor performance.
- Automatic isolation of a failed propulsion channel where appropriate.
- Reallocation of collective and attitude authority across healthy rotors.
- Pilot/vehicle warning hierarchy that distinguishes degraded operation from immediate emergency.
- Independent emergency power for flight-critical control functions.
- Hardware-in-the-loop and fault-injection testing before flight.

A critical design principle is that fault reconfiguration must not create a new unstable condition. Every claimed failure case therefore requires a defined control-law state, actuator authority assessment, structural load assessment, and flight-test envelope.

12. Structural and Load-Path Philosophy

The primary structure is organized into multiple load paths around the central capsule. Rotor loads, engine loads, landing loads, and emergency loads are not assumed to follow the same path.

- Rotor stations have dedicated local load paths into the primary frame.
- Capsule loads are transferred through controlled interfaces and dampers.
- Landing loads are routed through landing structures and energy absorbers before reaching the capsule.
- High-consequence joints receive dedicated inspection and test requirements.
- Structural sizing will use limit and ultimate load cases and must account for load redistribution.

EASA's CS-27 framework states that rotorcraft strength requirements are expressed in limit and ultimate loads and, unless otherwise provided, uses a 1.5 factor of safety. The eventual certification basis will depend on the aircraft category and certification authority selected for the product.

This document does not claim compliance. It uses certification principles as design-development guidance only.

13. Emergency Recovery Concept

Emergency recovery is intentionally treated as a layered function rather than a single parachute or a single capsule mechanism.

Scenario	Primary response	Secondary response	Validation required
Single rotor failure	Distributed control reconfiguration	Power redistribution / landing mode	Simulation + flight test
Multiple rotor degradation	Emergency flight mode	Aircraft recovery system	Failure-injection analysis + test
Major fire	Detection, isolation, power reduction	Emergency recovery / landing	Fire tests + system tests
Severe structural event	Maintain capsule integrity	Aircraft recovery concept	Dynamic structural test
Water impact / ditching	Capsule survivability and egress	Recovery / flotation provisions	Water-impact test

The emergency recovery concept includes an aircraft-level parachute system as a potential additional layer. Exact canopy count, deployment altitude, reefing strategy, attachment loads, and certification credit are not frozen by this revision.

14. Fire and Thermal Safety

- Thermal sensors in engine, generator, fuel and high-energy zones.
- Fire detection with independent logic from propulsion control where practical.
- Fuel isolation valves and emergency shutoff.
- Fire-resistant barriers between hot machinery and occupied space.
- Suppression system sized after hazard analysis.
- Emergency power reduction and load shedding.
- Post-fire safe-state logic.

The safety philosophy is based on containment plus isolation. The goal is not to assume that a fire can always be extinguished instantly; it is to prevent a local event from propagating into a loss of the occupied capsule.

15. Water Survival and Ditching

The capsule is intended to provide a protected survival volume during a water-landing scenario. The concept includes structural protection, controlled flotation, protected egress, and preservation of critical emergency equipment.

- Watertight or controlled-leakage boundaries around the occupant volume.
- Flotation provisions sized after mass-property and stability analysis.
- Emergency egress accessible after defined attitudes.
- Independent emergency electrical power.
- Location and activation logic for survival equipment.
- Capsule-to-airframe interfaces designed to limit water-impact shock transmission.

Water-impact survivability is a major validation item. It cannot be established from conceptual geometry alone.

16. Manufacturing and Prototyping Strategy

The compact baseline is deliberately structured for staged physical development. The recommended development path is not to build the complete aircraft immediately.

- Phase 1: coupon and material tests for structural and energy-absorption elements.
- Phase 2: subscale rotor, motor, controller and thermal-management tests.
- Phase 3: full-scale capsule impact and damper test articles.
- Phase 4: integrated propulsion test rig with multiple rotor channels.
- Phase 5: unmanned integrated demonstrator.
- Phase 6: progressive flight-envelope expansion.
- Phase 7: certification-oriented conformity and compliance programme.

This staged approach is a major reason for selecting the 10-ton baseline. The demonstrator can validate the safety architecture without requiring the resources of the superseded very-large aircraft concept.

17. Verification and Validation Plan

Subsystem	Analysis	Bench test	Dynamic test	Flight test
Rotor / hub	Yes	Yes	Yes	Yes
Propulsion channel	Yes	Yes	Yes	Yes
Flight controls	Yes	HIL	Fault injection	Yes
Capsule structure	FEA	Component	Full-scale impact	Limited / integrated
Dampers	Model	Characterization	Full-scale	Integrated
Fire protection	Hazard analysis	Component	System fire test	As required
Emergency recovery	Simulation	Component	Deployment test	Flight demonstration
Water survival	Simulation	Tank tests	Water-impact test	As required

The verification programme should trace every safety requirement to a measurable verification method and evidence record. No investor-facing performance claim should be upgraded from target to demonstrated performance without test evidence.

18. Certification Strategy

The certification pathway is not fixed in Document 00. The aircraft's final category, operational concept, occupant capacity, maximum takeoff mass, and jurisdiction will determine the applicable certification basis.

EASA currently maintains CS-27 for small rotorcraft and CS-29 for large rotorcraft. Current EASA material also emphasizes structural loads, critical parts, and test substantiation.

For a new rotorcraft, structural substantiation may require analysis supported by component, subcomponent, or full-scale testing. EASA's CS-27 material explicitly describes the role of static and dynamic testing in demonstrating strength and deformation compliance.

The project should therefore maintain a certification-assumption register and avoid presenting this conceptual document as evidence of compliance.

19. Safety Assessment Framework

- Functional Hazard Assessment for flight-critical and occupant-protection functions.
- Preliminary System Safety Assessment for propulsion, flight control, fire protection, emergency recovery and capsule systems.
- Common-cause and common-mode failure analysis.
- Failure propagation analysis between fuel, hot machinery, electrical systems and capsule.
- Structural failure and load-path assessment.
- Human factors and emergency egress assessment.
- Fault detection and annunciation assessment.
- Verification traceability from requirement to test evidence.

The ten-layer architecture is not ten independent guarantees. Several layers can share common causes of failure. The safety case must therefore explicitly analyze common-mode vulnerabilities and dependencies.

20. Product and IP Positioning

The compact 10-ton platform is intended to become the primary demonstrator and commercial flagship baseline. The strongest differentiating theme is not merely 'a helicopter with a capsule'; it is an integrated architecture in which the capsule, structural isolation, energy absorption, distributed propulsion, fire containment, and emergency recovery are designed as one safety system.

- Core architecture: fixed safety capsule + distributed rotor system.
- Mechanical isolation and energy-absorption architecture.
- Fault-tolerant distributed propulsion/control architecture.
- Fire and energy containment architecture.
- Emergency recovery and survival architecture.
- Modular interfaces supporting future detachable-capsule derivatives.

Any patent strategy should be developed from verified claimable mechanisms and carefully separated from generic aerospace features. This document is a technical baseline, not a legal opinion on patentability or freedom-to-operate.

21. Preliminary Risk Register

Risk	Impact	Current treatment	Next closure action
Mass growth above 10 t	High	Growth reserve retained	Supplier quotes + weighed hardware
Installed power higher than target	High	Propulsion trade study open	Rotor CFD + system efficiency model
Damper stroke insufficient	High	Multiple layers specified	Dynamic impact testing
Capsule structure too heavy	Medium/High	Dedicated mass allocation	FEA + material coupons
Common-mode control failure	High	Distributed architecture	SSA + HIL fault injection
Fire propagation	High	Containment + isolation layers	Fire hazard analysis + tests
Emergency recovery load exceedance	High	Recovery concept retained	Deployment simulation + structural test
Water-impact instability	High	Capsule survival concept	Water-impact modelling + test
Thermal management mass growth	Medium	Dedicated allocation	Thermal model + rig test

22. Open Engineering Items Before Rev 01

- Freeze mission profile: payload, crew, cruise speed, altitude, range and endurance.
- Complete rotor aerodynamic model and define rotor RPM, blade geometry and control method for the 6.0 m rotors.
- Select candidate prime mover and compare turboshaft, turbogenerator and hybrid-electric architectures.
- Close full power budget including realistic hover and climb margins.
- Develop first-order transmission, generator, motor and inverter thermal model.
- Replace planning mass allocations with component-level estimates and supplier data.
- Develop capsule structural load cases and damper stroke/force targets.
- Define emergency recovery architecture and determine whether aircraft-level recovery is feasible within the mass budget.
- Define water-impact and flotation requirements.
- Create a formal system safety architecture and failure-mode database.
- Map the final design to the applicable certification basis.

Rev 01 should be triggered by closure of the propulsion/power model, first structural sizing loop, first capsule energy-absorption model, and a controlled mass budget.

23. Design Rules for the 10-Ton Baseline

Rule 1

Do not allow the gross-mass target to become a paper-only number. Every major subsystem must carry a measurable mass allocation.

Rule 2

Do not claim safety credit for an emergency device before its deployment, structural, and human-survivability evidence exists.

Rule 3

Prefer passive mechanical safety layers where they can reduce dependence on software or electrical power.

Rule 4

Avoid single common-mode dependencies that can defeat multiple safety layers simultaneously.

Rule 5

Treat the capsule as a system, not simply as a stronger cabin.

Rule 6

Design interfaces for future modular derivatives without forcing the first demonstrator to carry the full complexity of the detachable version.

Rule 7

Maintain explicit separation between controlled baseline values, engineering targets, calculated estimates, and demonstrated performance.

24. Preliminary Performance Envelope

Document 00 intentionally avoids presenting a precise range, cruise speed, endurance, climb rate, or maximum altitude because those values depend strongly on the unresolved propulsion, rotor, aerodynamic, and mass models.

Metric	Current treatment
Gross mass	10,000 kg target
Hover thrust requirement	~98.1 kN total; ~14.01 kN per rotor
Illustrative 20% total thrust margin	~117.7 kN total; ~16.81 kN per rotor
Ideal induced hover power floor	~1.40 MW at ISA sea-level density (updated for 6.0 m rotors)
Installed power	TBD after aerodynamic and efficiency model
Range / endurance	TBD after mission and fuel-flow model
Cruise speed	TBD

The ideal hover figure is useful as a sanity check only. It is not a claim of achievable aircraft performance.

25. Investor and Due-Diligence Readiness

The 10-ton baseline is intentionally easier to communicate than the superseded very-large platform. The commercial story can be built around a demonstrable safety architecture, a compact physical demonstrator, and a staged validation programme.

- Clear controlled baseline.
- Visible safety architecture rather than a single headline feature.
- Prototype-friendly scale.
- Modular propulsion and control interfaces.
- Traceable engineering risks.
- Explicit distinction between concept, calculated estimate, and demonstrated result.

A professional investor package should pair this document with a one-page product sheet, a mass/power spreadsheet, a preliminary hazard analysis, a test roadmap, and evidence from physical demonstrators. The existence of a polished document alone should never be represented as proof of flight performance or certification.

26. Glossary

Term	Meaning
CDD	Conceptual Design Document.
T/W	Thrust-to-weight ratio.
HIL	Hardware-in-the-loop.
FEA	Finite element analysis.
CFD	Computational fluid dynamics.
LRU	Line-replaceable unit.
MTOM	Maximum take-off mass.
FHA	Functional Hazard Assessment.
SSA	System Safety Assessment.
Fixed capsule	Safety capsule that remains mechanically integrated with the aircraft and is not designed to separate in the baseline configuration.
Graceful degradation	Controlled reduction in capability following defined failures while preserving required control or landing functions.

27. Final Baseline Statement

Document 00 establishes the new Gen-3 compact flagship baseline: a 10-ton, seven-rotor rotorcraft with seven nominal **6.0 m** rotors, 1,600 L fuel capacity, a 17 m × 17 m × 4 m target envelope, and a fixed non-jettisonable safety capsule.

The central engineering proposition is layered survivability. The aircraft does not depend on one rescue mechanism. It combines structural containment, mechanical isolation, multiple damped load paths, fire protection, distributed propulsion, fault reconfiguration, emergency recovery concepts, and occupant energy management.

The baseline is deliberately conservative about claims. Quantitative performance, mass, power, recovery performance, and certification compliance remain subject to analysis and test. The next revision should close the propulsion and mass budgets first, because those two disciplines drive the feasibility of nearly every downstream system.

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END OF DOCUMENT 00